End-to-End Actuarial Policy Persistency Modelling

An actuarial and statistical framework for policy lapse analysis, risk ranking, model validation, and retention targeting

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Executive Summary

Background

Policy persistency is an important consideration in life insurance because lapse reduces the expected duration of premium inflows and can affect portfolio stability, profitability, and the assumptions used in actuarial analysis. Identifying where lapse risk is concentrated can therefore help insurers assess persistency experience and consider whether retention resources should be targeted rather than applied uniformly across the portfolio.

This project develops an end-to-end actuarial framework for analysing and ranking life insurance lapse risk using 10,000 synthetic policyholder records. The analysis combines data-quality assessment, actuarial assumption development, feature engineering, exploratory data analysis, persistency analysis, logistic regression, model validation, and portfolio risk segmentation.

A critical interpretation boundary applies throughout the project: the lapse outcome is synthetically generated from predefined actuarial assumptions rather than observed insurer experience. The analysis therefore demonstrates the modelling workflow and its ability to recover relationships embedded in the simulated portfolio; it does not establish empirical lapse rates or causal relationships applicable to a real insurer.

Analytical Approach

The analysis follows the sequence:

Data Quality → Actuarial Assumptions & Feature Engineering → Exploratory Analysis → Persistency Analysis → Predictive Modelling → Validation → Risk Segmentation → Actuarial Interpretation

Logistic regression is used as the primary predictive model because its coefficients can be translated into odds ratios and therefore provide an interpretable framework for examining relative lapse risk. Four specifications are compared: a null model, a raw-variable model, a business-engineered model, and a simplified final model.

The analysis then evaluates discrimination, classification behaviour, calibration, cross-validation, and multicollinearity before considering how predicted risk could be used for portfolio segmentation and retention prioritisation.

Key Findings

The portfolio contains 354 lapsed policies and 9,646 active policies, corresponding to an overall lapse rate of 3.54%. This relatively rare lapse outcome makes overall accuracy a poor standalone measure of model usefulness: a model predicting every policyholder as active would achieve approximately 96.5% accuracy while identifying no lapses.

Exploratory analysis identifies several material differences in lapse experience. Lapse rates are highest among the youngest policyholders, declining from 5.6% for ages 18–29 to 1.2% for ages 60+. Term Life has the highest observed lapse rate among the four products at 4.3%, compared with 1.7% for Whole Life. Current smokers also exhibit elevated lapse incidence at 5.4%, compared with 3.5% among non-smokers. These patterns are consistent with the actuarial assumptions used to construct the synthetic outcome and provide candidate variables for multivariable modelling.

The analysis also identifies evidence supporting affordability and duration as candidate risk characteristics. Lapsed policyholders have lower mean age, income, coverage and premium than active policyholders, while the premium-to-income ratio provides a more meaningful representation of relative premium burden than premium alone. Shorter policy durations are associated with higher lapse propensity, motivating the retention of continuous policy duration in the modelling framework.

Model Results

The final logistic regression achieves a test AUC of 0.639, compared with 0.643 for the raw-variable model. The final specification therefore does not materially improve discrimination, but it achieves broadly comparable predictive performance with a more parsimonious specification.

The model provides interpretable evidence of relative risk. Holding other model variables constant, policyholders aged 60+ have approximately 72% lower estimated lapse odds than the youngest reference group. Current smokers have approximately 97% higher estimated lapse odds than non-smokers, while Whole Life policyholders have approximately 59% lower estimated lapse odds than the relevant reference category. These coefficients describe associations within the synthetic portfolio rather than causal effects.

The conventional 0.50 classification threshold is unsuitable for this rare-event problem because it produces effectively no sensitivity to the lapse class. The analysis therefore places greater emphasis on risk ranking and lift than on binary classification accuracy. This distinction is central to the proposed actuarial application: the model is better viewed as a tool for prioritising relative lapse risk than as a highly accurate individual lapse classifier.

Actuarial and Business Interpretation

The strongest practical application of the framework is portfolio segmentation. Rather than treating all policyholders as equally likely to lapse, an insurer could use predicted risk to identify groups with comparatively elevated lapse incidence and investigate whether retention resources should be concentrated in those segments.

Potential applications include prioritising early-duration policies, identifying policyholders with relatively high premium burden, and segmenting retention activity according to policy and behavioural characteristics. However, a risk score alone does not demonstrate that intervention will prevent lapse. The economic value of a retention programme would require information on customer value, intervention cost, and treatment effectiveness, followed by controlled testing of the intervention.

The framework also has qualitative relevance to actuarial assumption-setting because segmented lapse-risk information could potentially complement portfolio-level persistency assumptions. Any such application would require validation against genuine insurer experience and appropriate actuarial governance.

Overall Assessment

The project demonstrates an end-to-end approach to actuarial predictive modelling rather than simply fitting a statistical model. Its principal result is not a claim of high predictive accuracy. Instead, it demonstrates how actuarial assumptions can be translated into measurable features, how those relationships can be tested using an interpretable model, how rare-event performance should be evaluated, and how model output can be translated into portfolio-level risk segmentation.

The principal limitation is the synthetic nature of the data and lapse outcome. Consequently, the results should be interpreted as a demonstration of methodology and analytical reasoning rather than as evidence of real-world lapse behaviour. External validation using observed policy histories and genuine lapse dates would be required before the framework could support operational retention decisions or production actuarial assumptions.

1. Introduction & Insurance Context

1.1 Policy Persistency and Lapse

Policy persistency describes the extent to which insurance policies remain in force over time, while lapse represents termination of a policy due to non-payment or cessation of coverage before the end of its intended duration.

From a modelling perspective, lapse can be considered as a binary outcome, whether a policyholder is classified as active or lapsed, while persistency analysis introduces an additional temporal perspective by considering how policy status varies with policy duration.

This project therefore considers policy persistency from two complementary perspectives:

- ***Lapse-risk modelling:*** estimating the relative likelihood that a policyholder belongs to the lapsed group.
- ***Persistency analysis:*** examining how lapse behaviour varies across policy duration and other policyholder characteristics.

The distinction is important because a predictive model answers primarily whether lapse risk differs between policyholders, whereas time-oriented persistency analysis considers when lapse-related patterns occur.

1.2 Business Problem

From an insurance business perspective, the objective is not simply to produce a binary prediction for every policyholder.

A more useful question is:

Can policyholder and policy characteristics be used to identify groups with comparatively higher lapse risk, and can this information support more targeted retention analysis?

This distinction becomes particularly important when lapse is relatively infrequent.

A model that classifies almost every policyholder as active could achieve high overall accuracy while identifying very few actual lapses. For a retention application, such a model would have limited practical value.

The project therefore places emphasis on risk ranking and segmentation, alongside conventional binary classification metrics.

The eventual business interpretation is consequently:

Policyholder characteristics → Estimated lapse risk → Risk ranking / segmentation → Identify higher-risk groups → Prioritise retention resources

This does not imply that the model itself demonstrates that a particular retention intervention will reduce lapse. Rather, it demonstrates how predictive analytics could potentially be incorporated into a targeted retention framework.

1.3 Project Objectives

In summary, the project was designed to assess data quality; explore lapse behaviour across policyholder/policy characteristics; develop actuarially motivated features; develop and validate an interpretable lapse model; and translate outputs into risk segmentation and retention implications.

The project was designed with around five principal objectives.

Assess the Data. Evaluate the structure, completeness, consistency, and modelling suitability of the available synthetic life insurance dataset before undertaking predictive analysis.

Investigate lapse behaviour. Examine how lapse rates vary according to demographic, financial, health, household, and policy characteristics.

Develop and validate an interpretable lapse-risk model. Use logistic regression to estimate lapse probability and evaluate the resulting model using discrimination, classification, calibration, multicollinearity, cross-validation, and risk-ranking diagnostics.

Translate statistical results into actuarial and business implications. Interpret model coefficients and risk patterns in an insurance context and consider how the resulting risk segmentation could potentially support targeted retention strategies.

1.4 Analytical Scope

The project deliberately focuses on an interpretable logistic-regression framework rather than attempting to maximise predictive performance through increasingly complex machine-learning algorithms.

This choice reflects the project's actuarial orientation.

The modelling workflow therefore progresses through alternative specifications rather than immediately fitting a single final model:

Null Model → Raw-Variable Model → Business-Engineered Model → Simplified Final Model

This allows the modelling decisions to be evaluated progressively rather than treating the final specification as an unexplained black box.

1.5 Interpretation Boundary

A fundamental consideration throughout this project is the nature of the underlying data.

The dataset contains 10,000 synthetic life insurance policyholder records, and the lapse outcome is generated using predefined assumptions relating characteristics such as age, policy duration, premium burden, smoking status, health, policy type, and household characteristics to lapse propensity.

Consequently, the project should not be interpreted as an empirical study of actual policyholder behaviour.

Instead, its purpose is to demonstrate whether an actuarially motivated modelling workflow can:

- Translate assumptions into an analytical target;
- Identify relationships embedded within the simulated data;
- Construct an interpretable predictive model;
- Evaluate model behaviour appropriately; and
- Translate the resulting analysis into potential insurance applications.

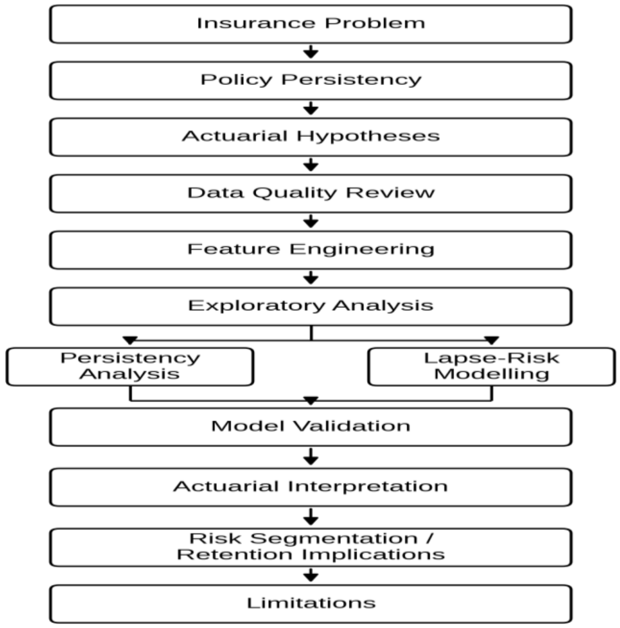
The existing project documentation explicitly establishes this boundary and notes that the model should not be interpreted as discovering real-world lapse drivers.

This distinction will be maintained throughout the report. Where the analysis identifies an association, it will be described as an association within the synthetic dataset, rather than as evidence of a causal relationship in real insurance portfolios.

1.6 End-to-End Analytical Framework

Section 2 begins by examining the quality and structure of the underlying dataset.

The complete project can therefore be summarised as:



2. Data & Data Quality Assessment

The first analytical stage is to establish whether the dataset is sufficiently well structured and internally consistent to support subsequent exploratory and predictive analysis. Rather than beginning immediately with lapse modelling, the raw dataset is inspected first so that data-quality issues can be identified before any transformation or modelling decisions are made. The R workflow explicitly treats this as Checkpoint 1 — Data Quality Assessment.

2.1 Dataset Overview

The project uses the Life Insurance Customer Retention Dataset, consisting of 10,000 synthetic policyholder records and 14 raw variables.

The variables represent several dimensions of a life insurance policyholder's profile:

- demographic characteristics;
- household characteristics;
- financial characteristics;
- health and lifestyle characteristics;

insurance product characteristics; and

- policy information.

The dataset also contains identifying and descriptive fields that require different treatment from predictive variables.

Table 1. Raw Dataset Structure

Category	Variables	Role
Demographic	Age, gender, marital status	Candidate predictors

Household	Number of dependents	Candidate predictor
Financial	Annual income, coverage amount, monthly premium	Candidate predictors / feature engineering
Health & lifestyle	Health status, smoking status	Candidate predictors
Policy	Policy type, policy start date	Candidate predictors / feature engineering
Descriptive	Occupation	Excluded from modelling
Identification	Customer ID, name	Excluded from modelling

The project's data dictionary classifies policy_start_date as the source variable from which policy duration is derived, while annual income and monthly premium contribute to the engineered premium-to-income ratio. Customer ID and name are retained in the raw dataset but excluded from predictive modelling.

2.2 Data Quality Assessment

The raw dataset was assessed before cleaning or feature engineering across four principal areas:

- dataset dimensions and variable types;
- missing values;
- duplicate records; and
- distribution and structural checks of numeric and categorical variables.

This sequencing is deliberate: the raw extract is preserved before any transformations so that subsequent decisions remain traceable and reproducible.

2.3 Missing Values

No missing values were identified across any of the 14 variables.

The assessment returned:

Total missing values: 0

Consequently, no imputation or missing-value deletion procedure was required.

Data-quality implication

Because there were no missing observations, the modelling dataset does not require an imputation strategy. Attention can instead be focused on whether individual variables are appropriate representations of policyholder characteristics for predictive modelling.

2.4 Duplicate Records

Two forms of duplication were assessed:

- fully duplicated rows; and
- duplicated customer_id values.

Both checks returned zero duplicates.

The analysis therefore treats each record as representing a unique policyholder within the assumptions of the dataset.

2.5 Numerical Variable Characteristics

The initial descriptive assessment provides the following summary:

A notable feature is the difference between the mean and median for several financial variables. For example, annual income has a mean of approximately $85,933 compared with a median of $76,000, while coverage amount has a mean of approximately $367,903 compared with a median of $280,000.

This indicates right-skewed financial distributions, which is relevant when interpreting measures of affordability and when deciding whether raw financial variables provide the most meaningful representation of policyholder behaviour.

The project subsequently investigates this issue through feature engineering rather than automatically assuming that the raw premium variable is the most appropriate affordability measure.

2.6 Categorical Variable Structure

The principal categorical variables contain the following distributions:

These distributions provide useful initial context for later modelling. In particular, the relatively small number of Variable Life observations and the placeholder categories require consideration when categorical predictors are prepared for regression.

2.7 Data Quality Issues Requiring Treatment

Although there are no missing or duplicated observations, three modelling-related issues were identified.

2.7.1 _RARE_ Categories

The gender variable contains 34 _RARE_ observations, while health status contains 36 _RARE_ observations.

These are placeholder categories rather than meaningful substantive levels.

They therefore require treatment before categorical predictors are incorporated into the final modelling dataset.

2.7.2 Occupation

Occupation is stored as a free-text field with inconsistent spelling and high cardinality.

Rather than attempting to force this variable into a categorical regression framework, the project excludes occupation from predictive modelling while retaining it in the original raw dataset.

This is a modelling decision rather than simply a data deletion decision:

A variable can contain information without necessarily being suitable for direct inclusion in a particular statistical model.

The R workflow therefore preserves the raw field for reproducibility while excluding it from the modelling dataset.

2.7.3 Customer Name and Customer ID

Customer name and customer ID are not predictive characteristics of policyholder behaviour.

They are therefore excluded from modelling.

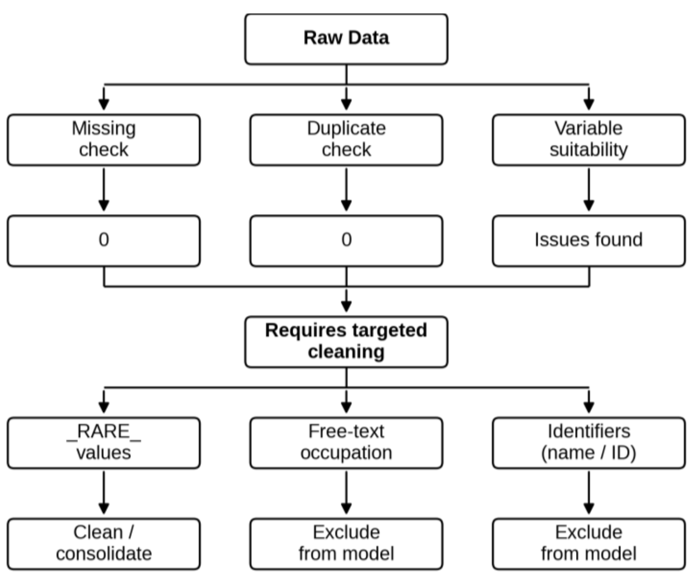
Importantly, the raw dataset is not overwritten. The modelling dataset is created as a separate cleaned object, allowing the original extract and subsequent transformation decisions to remain traceable.

2.8 Data Preparation Decision Framework

The data preparation decisions can be summarised as follows:

- Customer ID: identifier -> Exclude (no predictive meaning).
- Name: free-text identifier -> Exclude (no modelling relevance).
- Occupation: high-cardinality/free-text -> Exclude (sparse categories / limited robustness).
- Rare health category: sparse -> Consolidate (avoid unstable estimates).
- Policy start date: raw date -> Transform (derive policy duration).
- Premium/income: two raw variables -> Engineer ratio (represent affordability).

The data-quality assessment leads to the following treatment:



This produces a useful analytical distinction:

The dataset does not require *missing*-value or duplicate-record remediation, but it does require targeted variable-level cleaning and suitability decisions before modelling.

The project's Checkpoint 1 conclusion reaches the same overall assessment: the dataset is considered suitable for predictive modelling after targeted cleaning.

3. Actuarial Assumptions, Target Construction & Feature Engineering

3.1 Modelling Framework and Actuarial Assumptions

The purpose of this project is to demonstrate how policyholder characteristics can be translated into a structured lapse-risk modelling framework. Because the available portfolio is synthetic, the lapse outcome is generated from transparent assumptions designed to represent plausible actuarial relationships rather than being observed directly from insurer experience.

The assumptions were selected to reflect several broad mechanisms through which policyholders may differ in persistency:

Table 2. Factor Assumptions Structure

Factor	Assumed relationship with lapse propensity	Actuarial rationale
Age	Higher propensity among younger policyholders	Greater mobility and earlier financial/family formation stages
Smoking status	Higher propensity among current smokers	Risk profile and behavioural characteristics
Policy type	Higher propensity for Term Life; lower propensity for Whole Life	Differences in product structure, cash-value accumulation and surrender incentives
Health status	Higher propensity for less favourable health categories	Potential financial-strain and policyholder-characteristic differences
Marital status	Variation across household-status groups	Differences in household financial circumstances
Number of dependents	Higher propensity with more dependents	Greater competing financial priorities
Premium affordability	Higher propensity where premiums represent a larger share of income	Financial burden associated with maintaining cover
Policy duration	Higher propensity at shorter durations	Early-policy persistency differences

These assumptions provide the actuarial structure for the synthetic target. They are intended to create a portfolio in which lapse experience varies systematically across policyholder and policy characteristics, allowing the subsequent statistical analysis to evaluate whether those relationships can be recovered.

This distinction is fundamental to the interpretation of the project. Because the target is generated from predefined assumptions, the subsequent logistic regression should be viewed primarily as validation of the modelling workflow against known synthetic relationships, rather than as discovery of previously unknown real-world lapse drivers. The modelling results therefore demonstrate the analytical process while the synthetic nature of the outcome limits empirical interpretation.

3.2 Synthetic Lapse Target Construction

The target variable, Lapsed, is a binary classification:

- Active — policy remains in force;
- Lapsed — policy is classified as having lapsed.

The synthetic outcome is constructed using the actuarial relationships described above. Policyholder characteristics contribute to an underlying lapse propensity, which is then converted into the binary outcome.

The resulting portfolio contains 10,000 policies, comprising:

- 9,646 Active policies
- 354 Lapsed policies
- 3.54% overall lapse rate

This creates a pronounced class imbalance, with lapse representing only a small proportion of the portfolio.

The class distribution is not merely a descriptive feature of the dataset. It directly affects the later modelling strategy. Because a model can achieve high overall accuracy by predominantly predicting the majority Active class, subsequent evaluation must consider measures such as sensitivity, specificity, ROC-AUC and lift, rather than accuracy alone.

The target construction also creates an important interpretation boundary that is retained throughout the report:

The synthetic lapse outcome is suitable for demonstrating an actuarial modelling workflow, but it is not equivalent to observed insurer lapse experience.

The implications of this distinction are examined critically in Section 9.

3.3 Candidate Predictors and Feature Representation

The initial candidate predictors were selected to represent demographic, behavioural, product, affordability and duration characteristics available within the portfolio.

Most variables retain their original representation. Three areas received explicit consideration because their representation could materially affect model interpretation:

- ***Age:*** continuous age versus grouped age;
- ***Premium:*** raw monthly premium versus premium-to-income ratio;
- ***Policy duration:*** continuous policy duration as a potential retention-related predictor.

The feature-engineering process therefore focused on testing whether alternative representations provided sufficient evidence to justify additional modelling complexity, rather than engineering variables solely because they were intuitively plausible.

Age

Age was considered in both continuous and grouped form. The grouped version divides policyholders into five categories:

- 18–29
- 30–39
- 40–49
- 50–59
- 60+

Univariate logistic models produced AIC values of 3006.0 for continuous age and 3014.6 for age group. The continuous representation therefore provides the lower AIC and retains more information about the underlying age gradient.

Accordingly, continuous age was retained for modelling, while age groups were retained as an EDA and communication variable because grouped results are easier to interpret when presenting portfolio patterns.

This distinction is useful throughout the report: continuous age supports model estimation, while age groups support actuarial storytelling and segmentation.

Premium and affordability

Monthly premium was compared with an engineered premium-to-income ratio intended to represent affordability more directly.

Table 3. Univariate AIC values

Representation	AIC
Monthly premium	3062.7
Premium-to-income ratio	3064.7

The raw premium therefore performs marginally better in the univariate comparison. Nevertheless, the ratio was retained as a candidate for the multivariable engineered model because affordability may become informative after accounting jointly for income and other policyholder characteristics.

This provides a useful modelling principle: a variable should not be promoted solely because it has a plausible business interpretation, but a theoretically meaningful alternative can still be carried forward for formal multivariable comparison.

The later model comparison provides the final evidence on whether this engineering adds predictive value.

Policy duration

policy_duration_years is retained as a continuous predictor because policy duration has a direct persistency interpretation. Unlike age and premium, no alternative representation was developed at this stage.

Its inclusion allows the modelling framework to test whether the duration relationship observed during EDA provides incremental information once other policyholder characteristics are considered.

At the same time, duration requires particular care when used in the later persistency analysis. In this synthetic dataset, duration contributes to the construction of the lapse target, so it cannot be treated as an independently observed lapse event time. Section 5 therefore presents the Kaplan–Meier analysis only as an illustrative complement rather than genuine empirical time-to-lapse estimation.

3.4 Feature-Selection Decisions

The candidate-variable assessment can be summarised as follows:

Table 4. Feature-Selection Structure

Candidate	Alternative	Evidence	Decision
Age	Continuous vs grouped	Continuous age AIC = 3006.0 vs grouped age AIC = 3014.6	Continuous age retained
Premium	Monthly premium vs premium-to-income ratio	AIC = 3062.7 vs 3064.7	Both carried forward for multivariable comparison
Duration	Continuous duration	AIC = 3062.5	Retained as duration predictor
Age group	Grouped age	Useful for EDA interpretation	Retained as labelling/segmentation variable

This approach avoids treating feature engineering as an automatic improvement exercise. Instead, alternative representations are evaluated against evidence and their intended analytical role.

The resulting distinction between modelling variables and communication/EDA variables is particularly useful. Age group, for example, provides a clear way of communicating lapse gradients across customer cohorts, while continuous age retains greater information for the regression model.

3.5 Modelling Dataset and Interpretation Boundary

Following the assumptions and feature-engineering review, the dataset provides a structured set of candidate predictors for the modelling stages that follow.

The key analytical chain is:

Actuarial assumptions → Synthetic lapse propensity → Binary Lapsed / Active outcome → Candidate predictor representation → Exploratory analysis → Competing model specifications → Validation and risk segmentation

This structure is important because the later results should be interpreted considering how the target was constructed.

The synthetic assumptions provide a known relationship against which the modelling workflow can be assessed. The objective is therefore not to claim that the model has discovered causal relationships in real insurance experience, but to demonstrate that the analyst can:

- Translate actuarial reasoning into measurable variables;
- Construct and document a modelling target;
- Compare alternative feature representations;
- Make evidence-based modelling decisions; and
- Carry those decisions through to validation and portfolio application.

Section 4 now builds directly on these decisions by examining which relationships are actually visible in the resulting portfolio. Rather than repeating the assumptions, the EDA tests whether the synthetic portfolio exhibits the patterns that the assumptions were designed to create.

4. Exploratory Data Analysis & Lapse Behaviour

Exploratory data analysis was used to establish how lapse experience varies across the synthetic portfolio and to determine which characteristics warranted further investigation in the multivariable modelling stage. The purpose was therefore not to identify causal drivers or select the final model mechanically, but to connect the actuarial assumptions developed earlier with observable patterns in the data.

Four questions guided the analysis:

- Where is lapse risk concentrated across policyholder characteristics?
- Does financial burden appear related to lapse experience?
- Does policy duration differentiate persistency?
- Do product and policy characteristics provide useful segmentation?

The analysis identified several patterns that directly informed subsequent modelling decisions.

4.1 Portfolio Lapse Experience

The portfolio contains 10,000 policies, of which 354 are classified as lapsed and 9,646 as active, giving an overall lapse rate of 3.54%.

The low prevalence of lapse is important for the interpretation of the predictive modelling that follows. A model can achieve high overall accuracy simply by predicting the majority Active class for most observations while failing to identify lapse events. Consequently, the subsequent model assessment focuses on measures such as sensitivity, ROC-AUC, calibration and risk ranking rather than treating accuracy as sufficient evidence of usefulness.

This class imbalance also changes the business question. For retention purposes, the objective is not necessarily to classify every policyholder perfectly; it may be more useful to identify groups whose lapse incidence is sufficiently elevated to justify further investigation or targeted intervention.

4.2 Demographic and Policyholder Characteristics

Age

Age provides one of the clearest descriptive patterns in the portfolio. The observed lapse rate falls progressively across the age groups:

Table 5. Age Groups Structure

Age group	Policies	Lapse rate
18–29	1,789	5.6%
30–39	3,015	4.2%
40–49	2,366	3.1%
50–59	1,772	2.3%
60+	1,058	1.2%

The youngest group therefore experiences a lapse rate more than four times that of the 60+ group. This provides a strong descriptive basis for retaining age as a candidate predictor and motivates the use of age groups as an alternative representation of the relationship.

The pattern is consistent with the synthetic assumptions underlying the project. It should not, however, be interpreted as evidence that age itself causally determines lapse in real insurance portfolios.

Smoking Status

Smoking status also differentiates observed lapse experience:

- ***Current smoker:*** 5.4%
- ***Non-smoker:*** 3.5%
- ***Former smoker:*** 2.9%

Current smokers therefore exhibit approximately 1.5 times the portfolio-level lapse rate of non-smokers. The difference provides support for retaining smoking status in the multivariable model. Formal testing also provides evidence of an association between smoking status and lapse.

The result should again be interpreted as a characteristic of the simulated portfolio rather than evidence that smoking causes policy lapse.

Marital Status and Health Status

Marital status shows smaller but observable differences in lapse experience. Single policyholders have the highest lapse rate at 4.4%, compared with 3.3% for married policyholders and 2.4% for widowed policyholders.

Health status shows less consistent evidence. The observed lapse rate is highest among the small Poor-health group at 5.6%, but this category contains only 36 policies, making the percentage unstable relative to the much larger Excellent, Good and Fair groups. The formal chi-square test produces $p = 0.521$, providing limited evidence of an association between health status and lapse.

This illustrates why descriptive differences alone should not determine model inclusion. Both effect magnitude and statistical evidence need to be considered alongside multivariable predictive performance.

4.3 Product, Duration and Financial Characteristics

Policy Type

Policy type provides a more substantial segmentation pattern:

Table 6. Policy Type Structure

Policy type	Policies	Lapse rate
Term Life	5,579	4.3%
Variable Life	458	3.7%
Universal Life	2,053	3.1%
Whole Life	1,910	1.7%

Term Life therefore has approximately 2.5 times the observed lapse rate of Whole Life.

This difference makes policy type a useful candidate modelling variable and provides a potentially relevant portfolio segmentation dimension. However, the synthetic dataset does not contain sufficient real-world information to establish why the products differ. The interpretation is therefore limited to an observed association within the simulated portfolio. Formal testing provides strong evidence of differences across policy types ($\chi^2 = 33.2$, $p \approx 3 \times 10^{-7}$).

Policy Duration

Policy duration provides the direct link between EDA and persistency analysis. The exploratory results show higher lapse propensity among shorter-duration policies, consistent with the project's assumption that early policy years represent a period of greater lapse exposure.

Rather than replacing duration with broad categories in the predictive model, the analysis retains policy duration as a continuous variable. The duration bands remain useful for descriptive analysis because they make the portfolio pattern easier to communicate, while the continuous form preserves information for multivariable modelling.

The observed policy-duration range is relatively narrow, approximately 2.03 to 2.97 years. This means the analysis demonstrates behaviour within an early-to-mid policy window rather than providing evidence across conventional long-term persistency horizons.

Premium, Income and Affordability

The comparison of Active and Lapsed policyholders shows lower average values for several absolute financial variables among the lapsed group:

Table 7. Financial Variables of Active and Lapsed Structure

Variable	Active mean	Lapsed mean
Annual income	$86,125	$80,698
Coverage amount	$369,735	$317,994
Monthly premium	$158.78	$147.85
Number of dependants	1.28	1.16

Age also differs materially, with a mean age of 42.52 among active policyholders versus 37.83 among lapsed policyholders.

Absolute premium alone does not provide a complete affordability measure because a higher premium may simply reflect a higher-income policyholder or greater coverage. The analysis therefore considers the premium-to-income ratio as a relative measure of premium burden.

The ratio shows separation between the Active and Lapsed groups and is consequently carried forward as a candidate predictor. This is an actuarially more meaningful question than simply asking whether lapsed policyholders pay higher premiums:

Does the premium represent a different burden relative to the policyholder's income?

The subsequent modelling stage tests whether this relationship remains informative after controlling for other characteristics.

4.4 Statistical Evidence and Modelling Decisions

The exploratory analysis combines descriptive comparisons with formal statistical testing. For the principal categorical variables:

Table 7. Categorical Variable Findings

Variable	Test result	Interpretation
Smoking status	$p = 0.0149$	Evidence of association
Policy type	$p < 0.001$	Strong evidence of association
Marital status	$p = 0.0258$	Evidence of association
Health status	$p = 0.521$	Limited evidence

These tests provide screening evidence rather than final model-selection rules. A variable may show a group difference but contribute little incremental information once other predictors are included. Conversely, a variable with limited univariate evidence may still contribute conditionally within a multivariable model.

The main EDA findings therefore translate into the following modelling decisions:

Table 8. EDA Findings

Exploratory finding	Modelling implication
Lapse represents only 3.54% of observations	Evaluate sensitivity, AUC and lift rather than accuracy alone
Younger policyholders have higher lapse rates	Retain age as a candidate predictor
Shorter-duration policies show higher lapse propensity	Retain continuous policy duration
Premium burden differs between groups	Evaluate premium-to-income ratio
Current smokers have higher lapse rates	Retain smoking status
Policy types differ materially	Retain policy type
Marital groups differ in lapse rate	Retain as a candidate characteristic
Health-status evidence is weak	Retain initially but allow multivariable modelling to determine incremental value
Financial variables are right-skewed	Interpret scale and variable representation carefully

This demonstrates the role of EDA in the overall modelling framework: EDA informs the model; it does not dictate the final specification.

4.5 EDA Interpretation

Several relationships are sufficiently strong to provide a clear analytical narrative. Lapse is concentrated more heavily among younger policyholders, shorter-duration policies, current smokers and certain product types. Relative premium burden also provides a more economically meaningful representation of affordability than absolute premium alone.

However, the magnitude of a descriptive difference should not automatically be interpreted as predictive importance. The subsequent logistic regression is required to determine whether these relationships remain after controlling for other characteristics and whether they improve the model's ability to rank lapse risk.

Most importantly, the EDA establishes relationships within the synthetic portfolio only. It does not establish that any characteristic independently causes lapse, that the observed patterns would persist in another insurer's portfolio, or that a statistically significant relationship will necessarily improve predictive performance.

The next sections therefore move from descriptive evidence to two complementary analytical questions: how lapse behaviour varies over policy duration and how policyholder characteristics can be combined into a multivariable lapse-risk model.

5. Policy Persistency Analysis

5.1 Duration and Persistency Patterns

Exploratory analysis in Section 4 identified a relationship between policy duration and lapse experience, with shorter-duration policies exhibiting higher lapse propensity. This section examines the same relationship from a persistency perspective by considering how the probability of remaining in force changes across policy duration, rather than only whether a policy is ultimately classified as lapsed.

This distinction is important. The logistic regression developed in Section 6 models the probability of the binary outcome Lapsed versus Active. A persistency analysis instead considers the progression of policies through time and is therefore conceptually closer to the actuarial analysis of policy duration and survival.

For the synthetic portfolio, the observed policy-duration range is approximately 2.03 to 2.97 years. The illustrative Kaplan–Meier analysis therefore evaluates retention only within this relatively narrow window rather than at conventional 1, 3, or 5-year actuarial persistency points.

The estimated retention curve declines over the observed window:

Table 9. Estimated Retention Curve

Duration	Illustrative retention	95% confidence interval
2.13 years	99.6%	99.5%–99.7%
2.50 years	97.8%	97.5%–98.1%
2.87 years	93.2%	92.4%–94.1%

Within this synthetic portfolio, the decline indicates increasing lapse incidence as duration progresses through the observed window. However, the magnitude of these retention estimates should not be interpreted as empirical insurer persistency rates because the underlying lapse outcome is synthetic and policy duration was itself used in constructing that outcome.

The more useful contribution of this analysis is therefore demonstrating the distinction between a binary lapse model and a time-oriented persistency framework.

5.2 Persistency by Policy Type

Policy type provides a clear example of why persistency can be examined at a portfolio-segment level. The Kaplan–Meier curves show differences in the illustrative retention patterns across Term Life, Universal Life, Variable Life and Whole Life policies.

The corresponding log-rank test provides strong evidence that the persistency curves differ across policy types:

$$\chi^2 = 33.2,\ 3 \text{ degrees of freedom},\ p \approx 3 \times 10^{-7}.$$

The observed contribution to this test is particularly pronounced for Term Life and Whole Life. Term Life contains 241 observed lapse events against 192.7 expected, whereas Whole Life contains 32 observed events against 68.1 expected. This is consistent with the descriptive EDA finding that Term Life has higher lapse incidence than Whole Life.

The result demonstrates how persistency analysis can identify product-level differences in lapse experience that may be relevant for actuarial monitoring. In an operational setting, an insurer could use equivalent analysis to monitor whether particular products exhibit persistency deterioration relative to other products or to established assumptions.

However, the current analysis does not establish a real product-specific persistency experience because the event time is not independently observed.

5.3 Interpretation and Event-Time Limitation

The Kaplan–Meier curves in this project must be treated as illustrative rather than genuine time-to-lapse estimates.

In a conventional survival analysis, the event time would be independently observed: for example, the actual date on which a policyholder surrendered or otherwise lapsed their policy. Here, policy_duration_years is an input used, together with other policyholder characteristics, to construct the synthetic Lapsed outcome. There is therefore no independently observed lapse date in the dataset.

This creates an important interpretation boundary. The curves demonstrate how the duration-linked assumptions embedded in the synthetic portfolio manifest as a persistency pattern, but they should not be presented as evidence of an insurer's actual lapse hazard or as an empirical estimate of time-to-lapse.

This distinction does not make the analysis irrelevant. Instead, it demonstrates the complementary roles of the two modelling approaches:

Table 10. Analysis Framework Structured

Aspect	Persistency analysis	Logistic regression
Primary question	When / how does retention change?	Who is more likely to lapse?
Outcome	Time-to-event framework	Binary lapse outcome
Main output	Retention / survival pattern	Estimated lapse probability and relative risk
Portfolio use	Persistency monitoring	Risk ranking and segmentation
Status in this project	Illustrative	Primary validated predictive model

A genuine time-to-lapse study would require independently observed lapse dates and appropriate exposure/censoring information. A Cox proportional hazards model or another survival framework could then estimate how policyholder characteristics affect lapse hazard directly. This is a natural extension of the project but is intentionally outside the current modelling scope.

5.4 Persistency Analysis Takeaway

The persistency analysis adds a time-oriented perspective to the EDA and logistic modelling. The illustrative retention curve declines from 99.6% at 2.13 years to 93.2% at 2.87 years, while policy-type curves differ significantly in the synthetic portfolio.

The principal conclusion is methodological rather than empirical: persistency analysis can describe how lapse experience evolves over time, whereas logistic regression provides a framework for ranking policyholders according to their estimated probability of lapse.

For this project, the logistic regression remains the primary predictive model because the dataset contains a synthetic binary outcome rather than genuine independently observed lapse times. The persistency analysis is therefore best viewed as a complementary actuarial demonstration rather than as evidence of real-world policy-duration experience.

6. Predictive Model Development

6.1 Modelling Framework

The predictive objective is to estimate the probability that a policyholder is classified as Lapsed rather than Active. Logistic regression was selected as the primary modelling framework because it provides both a predictive probability and an interpretable relationship between policyholder characteristics and lapse odds.

The modelling process follows a deliberately staged progression:

Null model → Raw-variable model → Business-engineered model → Stepwise final model

This progression allows the analysis to distinguish between three questions:

- How much predictive information exists beyond the portfolio-wide lapse rate?
- Do the available raw variables provide useful incremental information?
- Can a more parsimonious specification retain comparable performance while improving interpretability?

Because the lapse outcome is synthetically generated from predefined assumptions, the modelling objective is not to discover previously unknown real-world lapse drivers. Instead, the model evaluates whether the statistical framework can recover and combine relationships embedded in the simulated portfolio. This interpretation boundary remains important when considering the coefficient estimates.

The dataset was divided into 7,001 training observations and 2,999 test observations, with lapse rates of approximately 3.54% and 3.53%, respectively. This preserves the rare-event structure between the development and held-out samples.

6.2 Model Development Progression

Four specifications were compared using training-set information criteria and held-out test-set performance.

Table 11. Model Development Performance Results

Model	Specification	Parameters	AIC	BIC	Test AUC
Model 0	Null	1	2145.9	2152.8	0.500
Model 1	Raw variables	20	2115.4	2252.5	0.643
Model 2	Business engineered	22	2119.2	2270.0	0.635
Model 3	Stepwise final	11	2103.8	2179.1	0.639

The null model provides the expected baseline: without policyholder characteristics, discrimination is effectively random with a test AUC of 0.500.

Introducing the raw predictors produces a substantial improvement, increasing test AUC to 0.643 and reducing AIC from 2145.9 to 2115.4. This demonstrates that the available policyholder characteristics contain information useful for distinguishing higher- and lower-risk policies.

The business-engineered model does not improve this result. Its test AUC falls from 0.643 to 0.635, while both AIC and BIC are higher than those of the raw-variable model. The engineered affordability and age representations therefore do not provide evidence of additional predictive value at the overall model level.

This is an important modelling result rather than a failure of the feature-engineering exercise. It demonstrates that an intuitively attractive engineered variable should not automatically be retained simply because it has a plausible actuarial interpretation. Its incremental value needs to be demonstrated empirically.

The stepwise final model reduces the specification from 20 raw-variable parameters to 11, while maintaining a test AUC of 0.639. It also produces the lowest AIC (2103.8) and BIC (2179.1) among the four specifications.

The final model therefore sacrifices only 0.004 AUC relative to the raw-variable model while substantially reducing model complexity.

This provides the principal rationale for selecting Model 3:

The final model was not selected because it had the highest predictive performance. It was selected because it retained broadly comparable discrimination while providing a materially simpler and more interpretable specification.

6.3 Final Model Interpretation

The final model contains age group, smoking status, policy type, and premium-to-income ratio. The coefficient estimates provide interpretable evidence of relative lapse odds within the synthetic portfolio.

Table 12. Final Model Results

Predictor	Odds ratio	Interpretation
Age 40–49	0.622	Approximately 38% lower lapse odds
Age 50–59	0.410	Approximately 59% lower lapse odds
Age 60+	0.281	Approximately 72% lower lapse odds
Current smoker	1.974	Approximately 97% higher lapse odds
Universal Life	0.659	Approximately 34% lower lapse odds
Whole Life	0.412	Approximately 59% lower lapse odds
Premium-to-income ratio	1215.612*	Positive association; raw one-unit OR is difficult to interpret

The premium-to-income ratio is measured on a scale where a one-unit increase represents a very large change in the ratio, making the raw odds ratio unsuitable for direct business interpretation.

The age effects are particularly pronounced. Relative to the reference age group, the estimated odds of lapse fall progressively for older groups, with the 60+ category having an estimated odds ratio of 0.281. In other words, its estimated lapse odds are approximately 72% lower than the reference group, holding the other model variables constant.

Smoking status shows the opposite direction. Current smokers have an estimated odds ratio of 1.974, corresponding to approximately 97% higher estimated lapse odds than the reference non-smoker group, conditional on the other predictors in the model.

Policy type also remains informative after adjustment. Whole Life has an estimated odds ratio of 0.412, corresponding to approximately 59% lower lapse odds relative to the reference policy type. Universal Life has an odds ratio of 0.659, while the Variable Life estimate is not statistically distinguishable from the reference category at the 5% level.

These results provide a useful example of why multivariable modelling adds value to EDA. The descriptive analysis identifies differences between groups; the logistic model evaluates whether those differences remain associated with lapse after accounting for other included characteristics.

However, these coefficients should be interpreted as conditional associations within the synthetic portfolio, not causal effects. In particular, the model does not establish that changing smoking status, age, product type or affordability would itself cause lapse risk to change.

6.4 Model Selection and Actuarial Interpretation

The model-development process provides three important findings.

First, the raw predictors contain meaningful information beyond the portfolio-wide baseline, increasing held-out AUC from 0.500 to 0.643.

Second, the business-engineered specification does not improve predictive performance. Its AUC of 0.635 is slightly below the raw-variable model's 0.643, indicating that additional engineered variables did not translate into better held-out discrimination.

Third, the stepwise specification achieves a useful balance between performance and complexity. Reducing the model to 11 parameters lowers AIC to 2103.8 and BIC to 2179.1, while retaining an AUC of 0.639.

The selection therefore reflects an actuarial modelling principle of balancing predictive information, parsimony and interpretability rather than pursuing the maximum possible model complexity.

Importantly, policy duration does not appear in the final specification despite being retained as a candidate predictor following EDA. This is itself informative: a variable can demonstrate a descriptive relationship with lapse while providing insufficient incremental value once other predictors are considered simultaneously.

The final model should consequently be viewed as an interpretable risk-ranking framework, rather than as a high-accuracy individual lapse classifier. Its predictive suitability is assessed formally in Section 7, while its potential use for portfolio segmentation and retention prioritisation is examined in Section 8.

6.5 Model Development Takeaway

The modelling progression demonstrates that the main analytical challenge is not simply fitting a logistic regression but deciding which level of model complexity is justified by the evidence.

The raw-variable model provides the highest held-out AUC at 0.643, but the final 11-parameter model achieves almost identical discrimination at 0.639 with lower information criteria. The engineered model provides no incremental predictive benefit.

The resulting model therefore represents a defensible compromise between performance and interpretability. Its coefficients provide actionable relative-risk information, while the modest AUC indicates that the model should ultimately be used for risk ranking and segmentation rather than deterministic prediction of individual lapse events.

7. Model Validation & Performance Evaluation

Model validation assesses whether the selected logistic regression provides useful and reasonably stable information beyond the training sample. Because only 3.54% of policies lapse, the evaluation prioritises discrimination, sensitivity, calibration and risk ranking rather than overall accuracy alone.

The validation evidence leads to a consistent conclusion: the model provides modest discrimination and should be interpreted primarily as a risk-ranking tool rather than a strong individual-level lapse classifier.

7.1 Threshold-Based Classification Performance

At the conventional classification threshold of 0.50, the model achieves an apparent accuracy of 96.5%, with 100.0% specificity but 0.0% sensitivity for the lapse class.

Table 13. Threshold-Based Classification Performance

Metric	Threshold = 0.50	Youden threshold ≈ 0.04
Accuracy	96.5%	60.5%

Sensitivity	0.0%	62.3%
Specificity	100.0%	60.4%

The 96.5% accuracy at the 0.50 threshold is therefore misleading. It largely reflects the dominance of the Active class: predicting almost every policyholder as Active produces high overall accuracy while failing to identify any lapse events.

This result is important from an actuarial perspective because a retention model that identifies no lapsing policyholders would have little practical value, regardless of its headline accuracy.

The threshold analysis demonstrates the trade-off more clearly. A threshold of approximately 0.04, selected using Youden's J statistic, increases sensitivity to 62.3% while reducing specificity to 60.4%. Overall accuracy falls to 60.5%, but this is not necessarily a deterioration in business usefulness: the lower accuracy reflects the model identifying a substantially larger proportion of the minority lapse class.

The 0.04 threshold should not, however, be interpreted as a recommended production threshold. It is an illustrative statistical operating point. A real insurer would need to determine the threshold according to intervention capacity, customer value, the cost of unnecessary retention activity and the cost of missed lapse opportunities.

The threshold analysis therefore supports an important modelling conclusion:

For this portfolio, predicted probabilities should not be converted into binary lapse decisions using the conventional 0.50 threshold. Relative risk ranking is more informative than default classification.

7.2 ROC Discrimination

The final model achieves a test ROC-AUC of 0.639 on the held-out test set.

An AUC of 0.639 indicates modest discrimination: the model contains information that helps distinguish higher-risk from lower-risk policyholders, but the separation is far from perfect.

This result is consistent with the model-development comparison in Section 6. The raw-variable model achieved a slightly higher test AUC of 0.643, while the selected final model achieved 0.639 with substantially fewer parameters. The small difference of 0.004 AUC supports the earlier decision to favour the simpler specification.

The ROC curve is particularly useful here because it evaluates ranking performance across possible thresholds rather than relying on the inappropriate 0.50 classification threshold. The model's value therefore lies more naturally in ordering policyholders by relative predicted risk than in assigning a definitive Lapsed/Active label.

This distinction is important when interpreting the model's potential business use. An AUC of 0.639 would not justify describing the model as a highly accurate lapse predictor. It can, however, provide a basis for concentrating attention on segments with comparatively higher predicted risk, subject to further validation.

7.3 Calibration

Discrimination assesses whether higher-risk policies tend to receive higher predicted probabilities, whereas calibration considers whether predicted probabilities are broadly consistent with observed lapse frequencies.

The test-set calibration results show broadly increasing observed lapse rates across the predicted-risk distribution, although the relationship is not perfectly monotonic in every decile:

Table 14. Test-set Calibration Results

Risk decile	Mean predicted	Observed lapse
1	1.17%	1.33%
2	1.66%	0.67%
3	2.14%	2.33%
4	2.60%	3.00%
5	3.25%	3.67%
6	3.72%	2.67%
7	4.26%	6.00%
8	4.50%	4.00%
9	5.39%	5.33%
10	6.98%	6.35%

The highest-risk decile has an observed lapse rate of 6.35%, compared with approximately 1.33% in the lowest-risk decile, indicating meaningful separation across the risk distribution despite some local deviations between predicted and observed rates.

The Hosmer–Lemeshow goodness-of-fit test gives:

$$\chi^2 = 5.7118,\ df = 8,\ p = 0.6795.$$

The p-value provides no strong evidence of a lack of calibration under this test. However, this should be interpreted as supporting evidence rather than proof that the model is perfectly calibrated. Calibration should ultimately be assessed using additional diagnostics and, in a real portfolio, monitored against subsequent experience.

Overall, the calibration evidence is consistent with using the predicted probabilities primarily for relative risk stratification, while further validation would be required before treating them as reliable absolute lapse probabilities.

7.4 Cross-Validation and Multicollinearity

Cross-Validation

Ten-fold cross-validation provides an additional assessment of whether the model's ranking performance depends heavily on the training/test split.

The mean cross-validated ROC is 0.6203, compared with the held-out test AUC of 0.639.

Table 15. Cross-validation Results

Performance measure	Value
Held-out test AUC	0.639
10-fold CV ROC	0.620

The difference is approximately 0.019, so the cross-validated result is directionally consistent with the held-out evaluation. This provides some evidence that the model's modest discrimination is not solely an artefact of one train/test split.

The result should nevertheless be described as evidence of reasonable stability, not evidence of strong generalisation. Both measures remain in the approximately 0.62–0.64 range, so the model's predictive strength is still modest.

The cross-validation output also reports sensitivity of 1.000 and specificity of 0.000 for its classification summary. These values should not be interpreted as evidence of an excellent classifier because the cross-validation metric is primarily being used here to assess ROC-based ranking stability, while the rare-event class structure makes threshold-based summaries highly sensitive to the operating threshold.

Multicollinearity

Multicollinearity was assessed using variance inflation factors for the final model. The maximum VIF is approximately 1.3, substantially below conventional concern thresholds.

This indicates that the retained predictors do not exhibit material linear redundancy within the final specification. Consequently, there is limited evidence that unstable coefficient estimates are being driven by excessive multicollinearity.

This result supports the interpretability of the final model alongside its reduced parameter count.

7.5 Overall Validation Assessment

Table 16. Validation Evidence

Validation dimension	Result	Assessment
Accuracy at 0.50	96.5%	Misleading under class imbalance
Sensitivity at 0.50	0.0%	Unsuitable for lapse identification
Specificity at 0.50	100.0%	Reflects majority-class prediction
Sensitivity at ≈0.04	62.3%	More useful minority-class detection
Specificity at ≈0.04	60.4%	Trade-off for higher sensitivity
Test AUC	0.639	Modest discrimination
10-fold CV ROC	0.620	Consistent with modest ranking ability
Hosmer–Lemeshow p-value	0.6795	No strong evidence of poor calibration
Maximum VIF	1.3	No material multicollinearity concern

Taken together, the validation results do not support describing the model as a high-performing individual lapse classifier. Its AUC remains modest, and the default 0.50 threshold fails to identify any lapse events.

The more defensible interpretation is that the model provides a modest but potentially useful ranking signal. The similarity between test AUC and cross-validated ROC provides some evidence of stability, while the calibration results and low VIF support the internal coherence and interpretability of the specification.

The practical value therefore depends less on whether the model can predict every individual lapse correctly and more on whether its ranking can concentrate lapse experience into identifiable higher-risk segments. This question is examined directly through risk deciles and lift in Section 8.

The distinction is important:

Validation establishes that the model contains a modest amount of predictive information; it does not establish that every high-risk policyholder will lapse or that contacting those policyholders will prevent lapse.

The latter questions require portfolio segmentation, intervention testing and economic evaluation, which are considered separately in the following sections.

8. Risk Segmentation, Lift & Retention Strategy

8.1 Risk Segmentation and Lift

The validation results indicate that the final model provides modest discrimination rather than strong individual-level classification. Its practical value therefore depends on whether predicted probabilities can concentrate observed lapse experience within higher-risk segments.

To assess this, policies in the held-out test set were ranked by predicted lapse probability and divided into risk deciles. This approach is particularly appropriate for a portfolio with a 3.54% overall lapse rate, where the objective may be to prioritise a limited proportion of policies for further retention activity rather than classify every policyholder perfectly.

The highest-risk decile has a lapse rate approximately 1.8 times the portfolio average, providing evidence that the model concentrates lapse experience toward the upper end of its predicted-risk distribution.

This is the key practical implication of the model's modest AUC. An AUC of 0.639 does not imply that the model can reliably identify every individual who will lapse. It does indicate that higher predicted-risk policies tend, on average, to contain a greater concentration of lapse events.

The risk-ranking framework can therefore be represented as:

Policyholder data → predicted lapse probability → risk ranking → targeted segment → retention decision

This is preferable to using the conventional 0.50 classification threshold identified in Section 7, which produced 0% sensitivity. The model is more appropriately treated as a mechanism for prioritising relative risk than as a binary decision rule.

8.2 Characteristics of the High-Risk Segment

To examine what the model's ranking represents in portfolio terms, the top 20% of predicted-risk policies were isolated from the test set.

The segment contains 600 policies, representing 20% of the test sample. Its actual lapse rate is 5.8%, compared with 3.5% across the full test set.

Table 17. Policy Lapse Results

Policies	2,999	600
Actual lapse rate	3.5%	5.8%
Relative lapse incidence	1.0×	≈1.66×

Table 18. Policy Share of Segment

Policy type	Share of high-risk segment
Term Life	88.0%
Universal Life	8.8%
Variable Life	2.5%
Whole Life	0.7%

Smoking status is more diverse, with 69.5% non-smokers, 16.8% current smokers and 13.7% former smokers.

The composition demonstrates an important point about multivariable risk modelling: the high-risk segment is not simply a list of smokers or a single demographic group. Risk ranking reflects the combined contribution of the characteristics retained in the final model.

At the same time, the segment should not be interpreted as a group in which most customers are expected to lapse. An observed lapse rate of 5.8% means that approximately 94.2% of the segment did not lapse in the test data.

This distinction is critical:

A high relative risk does not imply a high absolute probability of lapse.

Consequently, the model should be used to prioritise attention, not to label customers as certain or near-certain lapsers.

8.3 Retention Strategy and Actuarial Decision Framework

The model creates a potential basis for targeted rather than blanket retention activity. Instead of contacting the entire portfolio, an insurer could rank policies according to predicted lapse risk and allocate retention resources to segments where the expected value of intervention is greatest.

Table 19. Decision Framework

Stage	Decision	Relevant information
1	Rank	Predicted lapse probability
2	Segment	Risk decile / high-risk group

3	Assess value	Customer value, future premium and policy economics
4	Assess intervention	Expected effectiveness and intervention cost
5	Prioritise	Expected net benefit
6	Test	Controlled retention intervention
7	Monitor	Lapse, persistency and financial outcomes

The model's predicted probability should therefore be treated as one input into the retention decision, rather than the decision itself.

For example, a high-risk policy with low expected customer value may not justify an expensive retention intervention, while a moderately high-risk policy with substantial future premium value could warrant greater attention. The appropriate threshold should consequently depend on the economics of the retention programme rather than being selected solely from a statistical metric.

The current dataset does not contain sufficient information to calculate an economically optimal threshold. It does not provide reliable estimates of customer lifetime value, intervention cost or the probability that an intervention successfully prevents lapse.

The appropriate next step would therefore be to estimate the expected value of intervention using real insurer data:

Expected intervention value ≈ value of prevented lapse × probability intervention succeeds − intervention cost

This framework moves the analysis from prediction toward decision support, while recognising that prediction alone does not demonstrate that an intervention will change customer behaviour.

Retention Actions

Once a high-risk segment has been identified, potential interventions could include:

- targeted policy reviews;
- affordability or payment-option discussions;
- proactive customer contact during early policy duration;
- product or coverage reviews where appropriate;
- tailored communication rather than portfolio-wide campaigns.

These should be treated as candidate interventions, not recommendations proven by this model. The project does not estimate treatment effects and therefore cannot establish which intervention would reduce lapse.

8.4 Monitoring and Implementation Considerations

A production retention model would require ongoing monitoring beyond the initial validation reported in Section 7.

Table 20. Dimensions Monitoring:

Monitoring area	Example measure	Purpose
Discrimination	ROC-AUC / ranking performance	Detect deterioration in risk separation
Calibration	Predicted vs observed lapse rates	Detect changes in probability reliability
Portfolio experience	Actual lapse rate by segment	Compare modelled risk with emerging experience
Business impact	Retention rate, intervention cost and value retained	Determine whether the model creates economic value

Monitoring should also examine whether the characteristics of the underlying portfolio change over time. Changes in product mix, customer demographics, premium structure, or economic conditions could alter the relationship between predictors and lapse.

For a real insurer, model monitoring would therefore need to connect statistical performance with actuarial experience and business outcomes. A model could maintain an acceptable AUC while becoming less useful economically if intervention costs increase or customer behaviour changes.

The current synthetic project does not provide the longitudinal experience required to demonstrate such production monitoring. It instead establishes the analytical framework that could be applied once genuine policy histories become available.

The central practical conclusion is therefore:

The model's strongest demonstrated use is prioritisation: concentrating limited retention resources on policyholders whose predicted risk is relatively high, while recognising that intervention effectiveness and economic value must be established separately.

9. Limitations, Critical Evaluation & Actuarial Considerations

The modelling framework demonstrates an end-to-end approach to policy lapse analysis, but the results must be interpreted within several important limitations. These limitations do not invalidate the analysis; rather, they determine what conclusions the project can reasonably support and what would be required before applying the framework to real insurer experience.

The most important limitations relate to the synthetic construction of the outcome, the modest predictive performance, model specification and generalisability, the distinction between prediction and causality, and the requirements for real-world deployment.

9.1 Synthetic Data and Outcome Construction

The most fundamental limitation is that the dataset and lapse outcome are synthetic. The lapse indicator was constructed using predefined actuarial assumptions linking policyholder characteristics to lapse propensity. Consequently, the modelling process is not discovering unknown relationships from independently observed insurance experience.

This creates a clear interpretation boundary:

The model validates whether the statistical workflow can recover relationships embedded in the simulated portfolio; it does not establish empirical lapse relationships for a real insurer.

This distinction is particularly important when interpreting the odds ratios reported in Section 6. For example, the model identifies substantially higher estimated lapse odds for current smokers and lower estimated lapse odds for older age groups. These results demonstrate that the logistic regression successfully captures relationships present in the simulated data, but they should not be interpreted as evidence that these characteristics independently cause policyholders to lapse in real insurance portfolios.

Circularity and recoverability

Because several predictors contribute to the construction of the synthetic lapse outcome, some degree of recoverability is expected. A model that identifies these predictors is therefore demonstrating successful recovery of the assumptions used to generate the target rather than discovering an entirely independent empirical pattern.

This is a deliberate feature of the project rather than an accidental modelling error. It makes the dataset suitable for demonstrating:

- actuarial assumption design;
- feature engineering;
- exploratory analysis;
- logistic regression;
- model validation;
- risk segmentation; and
- translation of model output into an actuarial decision framework.

However, it means that predictive performance should not be interpreted as evidence that the same model would achieve similar performance on genuine insurer data.

Binary outcome simplification

The target is also simplified to Lapsed versus Active.

Real policy persistency is more complex. Policies may surrender, lapse for non-payment, become paid-up, be reinstated, mature or otherwise change status. The binary target therefore compresses several potentially different policy behaviours into one outcome.

A real insurer analysis would ideally distinguish the relevant policy-status transitions and incorporate exposure and event timing. This would allow lapse assumptions to reflect not only whether a policy exits the portfolio but also when and why that transition occurs.

Lack of genuine event time

The same limitation affects the Kaplan–Meier analysis in Section 5. The policy-duration variable is available, but the dataset does not contain an independently observed date at which each lapse occurred. The resulting survival analysis is therefore illustrative rather than a genuine empirical time-to-lapse analysis.

A real persistency study would require policy inception dates, lapse dates and appropriate censoring/exposure information. This would permit estimation of actual duration-specific lapse or hazard rates rather than demonstrating the time pattern implied by the synthetic assumptions.

9.2 Predictive Performance and Reliability

The second major limitation is the model's modest predictive discrimination.

The final model achieves a test ROC-AUC of 0.639, while 10-fold cross-validation produces a mean ROC of approximately 0.620. These results indicate useful information beyond random ranking, but they do not support describing the model as a highly accurate individual-level lapse predictor.

This limitation is important because a model with modest AUC can still produce useful portfolio segmentation. The top-risk segment demonstrates higher lapse incidence than the portfolio average, but this should not be confused with accurate prediction of individual lapse events.

The model's performance should therefore be framed as:

Modest individual-level discrimination with potential portfolio-level risk-ranking value.

Class imbalance

Only 3.54% of policies lapse, creating pronounced class imbalance. This explains why the conventional 0.50 probability threshold produces approximately 96.5% accuracy but 0% sensitivity.

The result demonstrates that accuracy is not an adequate measure of model usefulness for this problem. A model that predicts nearly every policy as Active can achieve high accuracy while providing no meaningful lapse identification.

The project addresses this by examining threshold behaviour, ROC-AUC, calibration and lift rather than relying on accuracy alone.

Nevertheless, the imbalance remains a limitation. In real insurer data, modelling choices may need to consider larger portfolios, alternative sampling approaches, class-weighting strategies or specialised rare-event methods depending on the application.

Limited feature-engineering gain

The business-engineered model also provides an important negative result. Engineering affordability and age-related variables did not improve held-out discrimination: the raw-variable model achieved an AUC of 0.643, compared with 0.635 for the engineered specification.

This demonstrates that theoretically plausible feature engineering does not automatically improve predictive performance.

The result also suggests that the current dataset may contain relatively limited additional information beyond the variables already represented in the raw predictors. On real insurer data, additional variables could potentially provide more predictive signal.

Calibration

The calibration results provide no strong evidence of poor fit under the Hosmer–Lemeshow test (p = 0.6795) and observed lapse rates generally increase across higher predicted-risk groups.

However, this should not be interpreted as proof of perfect calibration. Calibration is portfolio- and time-dependent, and the synthetic nature of the target means that apparently reasonable calibration within this dataset does not establish that predicted probabilities would remain reliable on a future real portfolio.

External and temporal validation would therefore be required before treating predicted probabilities as production lapse assumptions.

9.3 Model Specification and Generalisability

The final model is deliberately parsimonious, but this introduces a trade-off between interpretability and the ability to capture more complex relationships.

The selected logistic regression assumes a particular functional structure. Relationships that are nonlinear, interactive or conditional on other policy characteristics may therefore be inadequately represented.

For example, the effect of affordability may differ according to:

- age;
- policy type;
- income level;
- policy duration; or
- coverage amount.

Similarly, the effect of policy type may interact with customer characteristics. The current model does not comprehensively investigate such interactions.

A more flexible modelling strategy could therefore potentially identify additional predictive structure. Candidate extensions include:

- interaction terms;
- spline-based nonlinear effects;
- generalised additive models;
- tree-based models;
- gradient boosting; and
- other machine-learning approaches.

However, greater predictive flexibility would introduce an important trade-off. More complex models may improve discrimination while making coefficient interpretation, governance and actuarial explanation more difficult.

The appropriate objective should therefore not be maximum predictive complexity, but an appropriate balance between predictive performance, stability, interpretability and business usefulness.

Omitted variables

The model is also constrained by the variables available in the synthetic dataset. Real lapse behaviour can potentially reflect factors not represented here, such as:

- payment history;
- premium payment method;
- policy servicing activity;
- claims or benefit utilisation;
- policy amendments;
- agent or distribution channel;
- customer engagement;
- changes in financial circumstances;
- macroeconomic conditions; and
- prior contact or retention interventions.

Omission of relevant variables can reduce predictive performance and may also distort the estimated contribution of included variables.

The modest AUC should therefore not necessarily be interpreted as evidence that lapse behaviour is intrinsically unpredictable. It may partly reflect the limited information available to the model.

Generalisability

The model is evaluated using a held-out test sample drawn from the same synthetic portfolio-generating process. This provides internal validation but does not constitute external validation.

A model that performs adequately within one synthetic population may perform differently when applied to:

- another insurer;
- another product portfolio;
- another geographic market;
- another distribution channel; or
- a different economic environment.

Differences in customer composition, product design, underwriting, premium structures and lapse definitions could all change the relationship between predictors and lapse.

External validation on independent insurer experience would therefore be necessary before making claims about generalisability.

9.4 Prediction, Causality & Intervention

A central limitation is the distinction between predictive association and causality.

The logistic regression estimates associations between policyholder characteristics and the probability of the synthetic lapse outcome. It does not establish that changing one of those characteristics would cause the probability of lapse to change.

For example, the finding that current smokers have higher estimated lapse odds does not imply that changing smoking status would reduce lapse probability by the corresponding amount. Similarly, the lower lapse odds associated with Whole Life policies do not demonstrate that moving a customer into a Whole Life product would cause their lapse risk to fall.

This distinction becomes particularly important when the model is considered for retention activity.

Prediction does not establish intervention effectiveness

The model identifies who appears relatively more likely to lapse, but it does not identify:

who can be successfully persuaded not to lapse.

These are different modelling problems.

A policyholder may have a high predicted lapse probability but be unaffected by a retention intervention. Conversely, a policyholder with moderate predicted risk might respond strongly to a particular intervention.

Therefore, using the model to contact high-risk policyholders without evaluating treatment effectiveness could result in unnecessary intervention costs without demonstrating that lapse has actually been prevented.

Intervention testing

A stronger real-world retention framework would combine the lapse-risk model with controlled intervention testing.

For example, eligible high-risk policyholders could be randomly assigned to:

a retention intervention group; and

- a suitable control group.

Subsequent lapse experience could then be compared to estimate the incremental effect of the intervention.

This would move the analysis from:

"Who is likely to lapse?"

towards:

"Who is likely to respond to a particular retention intervention?"

That second question is closer to an uplift or treatment-effect modelling problem and is outside the scope of the current project.

9.5 Governance and Real-World Deployment

Before a model of this type could be deployed operationally, it would require substantially more validation and governance than is possible within the current synthetic project.

Temporal validation and model drift

The current test set is a held-out sample from the same underlying synthetic population. A production model should additionally be tested against future policy experience.

Lapse behaviour can change over time because of economic conditions, product changes, customer behaviour, pricing, distribution strategy and other environmental factors.

A model could therefore experience performance deterioration even if the underlying statistical implementation remains unchanged.

Production monitoring should include:

- ROC-AUC and ranking performance;
- calibration;
- observed versus expected lapse rates;
- risk-segment stability;
- population characteristic drift;
- intervention outcomes; and
- financial/business impact.

Fairness and responsible use

Risk segmentation can also create governance concerns if characteristics associated with protected or sensitive groups are used inappropriately.

The current synthetic analysis is not sufficient to establish whether any predictor would be appropriate for operational retention decisions under a real insurer's legal, regulatory and governance requirements.

A production implementation would therefore require review of:

- variable eligibility;
- fairness and potential disparate impact;
- explainability;
- data governance;
- customer treatment;
- regulatory requirements; and
- appropriate human oversight.

This is particularly important if model outputs influence customer contact or product recommendations.

Economic validation

Predictive performance alone does not establish that a model creates economic value.

A retention programme would need to consider:

Expected value of intervention = value of prevented lapse × probability of successful intervention − intervention cost

The current dataset does not contain sufficiently reliable customer-value or intervention-effectiveness information to calculate this quantity.

Consequently, the project demonstrates a decision framework, rather than a proven positive return from retention activity.

Production validation

Before operational use, a real insurer would also need:

- independent validation;
- temporal or out-of-time testing;
- monitoring standards;
- documented model assumptions;
- appropriate governance approval;
- reproducible data and model pipelines; and
- ongoing review of performance and business outcomes.

The current project should therefore be understood as a development-stage analytical framework, not a production-ready lapse model.

9.6 Development Path and Critical Assessment

The limitations above also define the most valuable next steps for developing the project.

Replace synthetic outcomes with genuine policy experience

The most important improvement would be to use observed policy histories containing:

- policy inception date;
- lapse/surrender date;
- exposure period;
- reinstatement information;
- policy status;
- premium history; and
- customer and policy characteristics.

This would allow the modelling framework to move from recovering known synthetic relationships toward learning from genuine insurer experience.

Develop a genuine time-to-event framework

With observed lapse dates, the project could be extended using survival-analysis techniques such as:

- Kaplan–Meier estimation using genuine event times;
- Cox proportional hazards models;
- time-varying covariates; and
- duration-specific hazard modelling.

This would provide a more actuarially natural representation of persistency.

Test nonlinear and benchmark models

The logistic regression provides interpretability, but future development could compare it with more flexible approaches such as generalised additive models, random forests or gradient boosting.

The objective should be to determine whether additional complexity produces a material and stable improvement in out-of-sample performance, rather than simply achieving a better in-sample fit.

Move from prediction to intervention modelling

The current model addresses lapse propensity. A future retention study could investigate treatment response through controlled experiments and uplift modelling.

This would provide a stronger basis for answering the operational question:

Which customers should receive which intervention?

rather than simply:

Which customers are more likely to lapse?

Overall critical assessment

Despite these limitations, the project demonstrates several capabilities that are directly relevant to actuarial analytics.

It demonstrates the ability to:

- translate actuarial assumptions into a structured dataset;
- perform data-quality and exploratory analysis;
- engineer and evaluate candidate predictors;
- compare competing statistical specifications;
- interpret logistic-regression coefficients as odds ratios;
- recognise the implications of rare-event classification;
- validate discrimination and calibration;
- assess multicollinearity;
- segment a portfolio using predicted risk;
- distinguish statistical performance from business usefulness; and
- identify the additional evidence required before operational deployment.

The strongest feature of the project is therefore not the absolute predictive performance of the model. It is the analytical judgement demonstrated in interpreting what the model can and cannot support.

The final model achieves only modest discrimination, but the analysis does not attempt to conceal this behind headline accuracy. Instead, the model is reframed as a potential portfolio-level risk-ranking tool, while intervention effectiveness, causality, generalisability and economic value are explicitly identified as questions requiring additional evidence.

This is the appropriate boundary between a demonstration of actuarial predictive modelling and a production-ready insurance decision system.

10. Conclusion & Recommendations

10.1 Overall Findings

This project developed an end-to-end actuarial analytics framework for analysing life insurance lapse risk, combining exploratory analysis, policy persistency analysis, logistic regression, model validation and risk segmentation.

The synthetic portfolio contains 10,000 policies, of which 354 are classified as Lapsed, giving an overall lapse rate of 3.54%. The analysis identified meaningful differences in lapse experience across several policyholder and policy characteristics, particularly age, smoking status and policy type.

The modelling stage compared a baseline model with raw-variable, business-engineered and simplified specifications. The selected final model reduced the number of parameters while retaining broadly comparable predictive performance to the larger raw-variable model. Its held-out test AUC was 0.639, compared with 0.643 for the raw-variable specification, while the final model achieved lower AIC and BIC.

The final odds-ratio analysis identified several substantial associations within the synthetic portfolio. For example, current smokers had estimated lapse odds approximately 1.97 times those of the reference group, while the estimated odds ratios for the 50–59 and 60+ age groups were 0.410 and 0.281, respectively. Whole Life policies had estimated lapse odds of 0.412 relative to the reference policy type.

These estimates should be interpreted as associations within the synthetic portfolio, rather than causal effects or empirical insurer experience.

Model validation produced a more nuanced result. The model demonstrated modest discrimination, with a test AUC of 0.639 and a 10-fold cross-validated ROC of approximately 0.620. The conventional 0.50 classification threshold produced high apparent accuracy but failed to identify lapse events because of the portfolio's pronounced class imbalance. This demonstrated why accuracy alone is inappropriate for evaluating rare-event lapse prediction.

The model's principal practical value therefore lies in risk ranking rather than individual-level classification. The highest-risk segments contained a greater concentration of observed lapse experience, with the top 20% risk segment recording a 5.8% lapse rate compared with 3.5% across the test portfolio. The highest-risk decile achieved approximately 1.8× portfolio-average lapse incidence.

Taken together, the evidence suggests that the model can provide a useful analytical basis for portfolio segmentation and retention prioritisation, but it should not be presented as a highly accurate individual lapse predictor or as a production-ready retention system.

The project's most important conclusion is therefore not that the model predicts lapse perfectly. It is that an actuarial modelling workflow can transform policy-level information into an interpretable risk-ranking framework, while validation and critical evaluation determine the boundaries within which that framework can responsibly be used.

10.2 Recommendations for an Insurer

Table 21. Proposed Recommendation

Recommendation	Proposed application	Required evidence
Prioritise high-risk policies	Rank policies by predicted lapse probability rather than applying a blanket retention campaign	Stable out-of-sample risk ranking
Segment retention activity	Focus resources on higher-risk segments while considering customer value	Risk concentration and economic analysis
Use customer value alongside risk	Prioritise policies where the potential value of preventing lapse justifies intervention cost	Customer value and intervention-cost data
Test interventions experimentally	Compare retention outcomes between intervention and control groups	Controlled treatment testing
Monitor model performance	Track discrimination, calibration, observed lapse rates and segment stability	Ongoing policy experience
Review assumptions regularly	Re-estimate lapse relationships as portfolio experience changes	Temporal validation and experience analysis

The model should therefore form one component of an actuarial decision process, rather than automatically determining which customers receive retention interventions.

In particular, the 5.8% lapse rate observed in the top 20% risk segment means that the majority of policies in that segment still did not lapse. High predicted risk should consequently be interpreted as relative prioritisation, not certainty of lapse.

Before operational deployment, an insurer would also need to establish whether retention interventions actually change outcomes and whether the resulting financial benefit exceeds intervention costs.

10.3 IFRS 17 Actuarial Relevance

The analysis also illustrates how lapse assumptions can contribute to the wider actuarial process under IFRS 17.

Lapse behaviour can influence projected policy cash flows and therefore the valuation of insurance contracts. A segmented lapse framework can potentially support the development of more differentiated assumptions by policy characteristics rather than relying exclusively on a single portfolio-wide lapse rate.

However, the application demonstrated by this project is deliberately qualitative. The dataset is synthetic and does not contain a real insurer's cash flows, contractual boundaries, liability calculations or financial reporting data. The project therefore does not attempt to calculate an IFRS 17 liability or quantify a CSM/LRC impact.

The appropriate interpretation is instead:

policyholder characteristics → lapse assumptions → projected persistency → insurance cash flows → actuarial valuation

A real implementation would require observed lapse experience, appropriate exposure periods, assumption-setting methodology and integration with the insurer's wider valuation and financial-reporting framework.

This provides actuarial relevance without overstating what the current dataset can demonstrate.

10.4 Future Modelling & Actuarial Development

The most valuable next development would be to replace the synthetic target with genuine policy histories containing independently observed lapse dates and exposure information. This would address the project's most important limitation and allow the framework to move from validating a known synthetic relationship toward learning from actual experience.

Several extensions would then become possible.

Genuine survival modelling

Observed lapse dates would allow the persistency analysis to move beyond the illustrative Kaplan–Meier framework toward genuine time-to-event modelling. Cox proportional hazards models, time-varying covariates and duration-specific hazard models could be considered.

This would allow the analysis to answer not only:

"Which policies are more likely to lapse?"

but also:

"How does lapse hazard change throughout the policy lifetime?"

Nonlinear and benchmark models

The logistic regression could be compared with more flexible models, including generalised additive models and tree-based methods. The objective should not simply be to maximise predictive performance, but to determine whether additional complexity produces a material, stable and interpretable improvement over the logistic benchmark.

Dynamic lapse modelling

A richer dataset could incorporate changing policy and customer information over time, allowing the model to update risk as circumstances change rather than relying only on information available at one point in time.

Uplift and intervention modelling

The current model estimates lapse propensity. A future retention study could investigate treatment response using controlled intervention data and uplift modelling.

This would address the more commercially relevant question:

Which customers are most likely to be retained because of a particular intervention?

rather than simply identifying customers who are more likely to lapse.

External and temporal validation

Finally, a production-oriented model should be tested on independent and future experience to establish whether its predictive relationships remain stable across time, products and portfolios.

These extensions would transform the current project from an illustrative actuarial modelling framework into a more realistic policy persistency and retention analytics system.

10.5 Final Conclusion

This project demonstrates an end-to-end approach to actuarial lapse modelling: from defining a policy-level target and examining persistency patterns, through model development and validation, to portfolio risk segmentation and potential retention application.

The final model does not provide strong individual-level prediction. Its 0.639 test AUC is modest, and the project's synthetic construction prevents the results from being interpreted as empirical evidence of real-world lapse behaviour. These limitations are central to the interpretation of the work rather than weaknesses to be concealed.

The stronger finding is that the modelling framework can produce interpretable and potentially useful relative-risk information. The higher lapse concentration observed within the upper-risk segments demonstrates how predictive modelling could support more targeted actuarial analysis and retention prioritisation than a portfolio-wide approach.

For an insurer, the next step would not simply be to deploy this model. It would be to combine genuine policy experience, time-to-event information, customer value, intervention testing and ongoing model governance to determine whether the predicted risk translates into measurable persistency improvement and economic value.

Ultimately, the project demonstrates the analytical judgement required in actuarial data science: build the model, quantify its performance, interpret its results, understand its limitations, and translate its output into a decision framework without claiming more than the evidence supports.